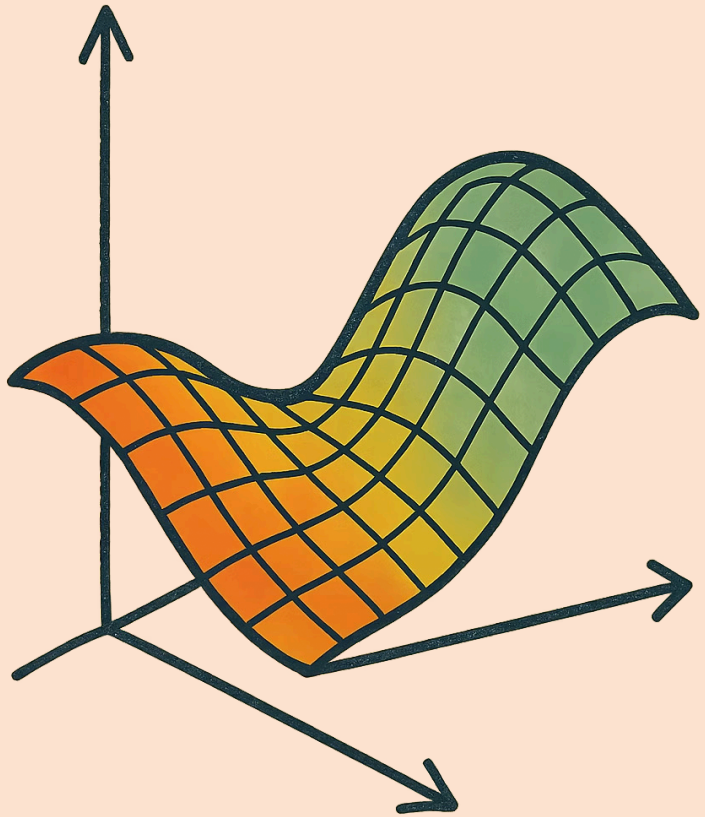

Multivariable Calculus: Differentiation

A B.Tech Engineering Mathematics journey — from single-variable foundations to gradients, directional derivatives, and real-world applications.

ENGINEERING MATHEMATICS

B.TECH CURRICULUM

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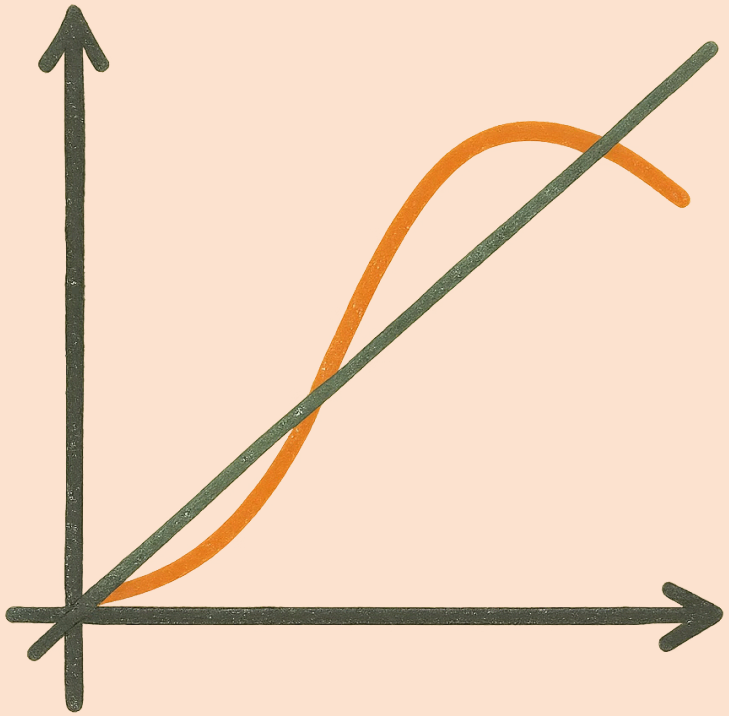

Recalling Single Variable Differentiation

Before extending to multiple variables, recall the core idea: the derivative measures **instantaneous rate of change**.

Definition

$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

Geometric Meaning

Slope of the tangent line at a point on the curve $y = f(x)$

Physical Meaning

Velocity = derivative of position; acceleration = derivative of velocity

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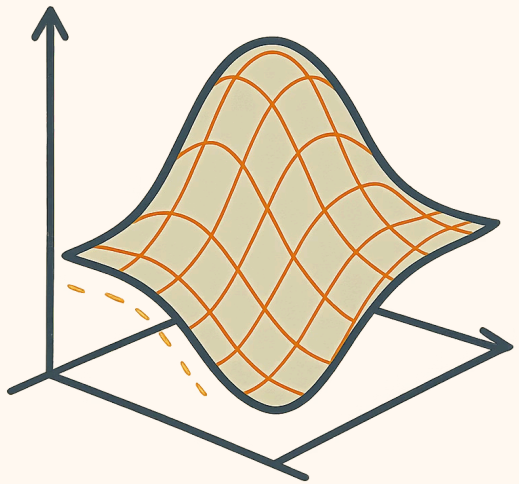
Introduction to Partial Derivatives



When a function depends on **multiple variables**, we differentiate with respect to **one variable at a time**, treating others as constants.

Notation: $f_x,\ f_y,\ \partial f/\partial x$ are all equivalent.

Partial w.r.t. x

$\frac{\partial f}{\partial x} = \lim_{h\to 0} \frac{f(x+h,y)-f(x,y)}{h}$

Partial w.r.t. y

$\frac{\partial f}{\partial y}$ — hold x constant, vary y

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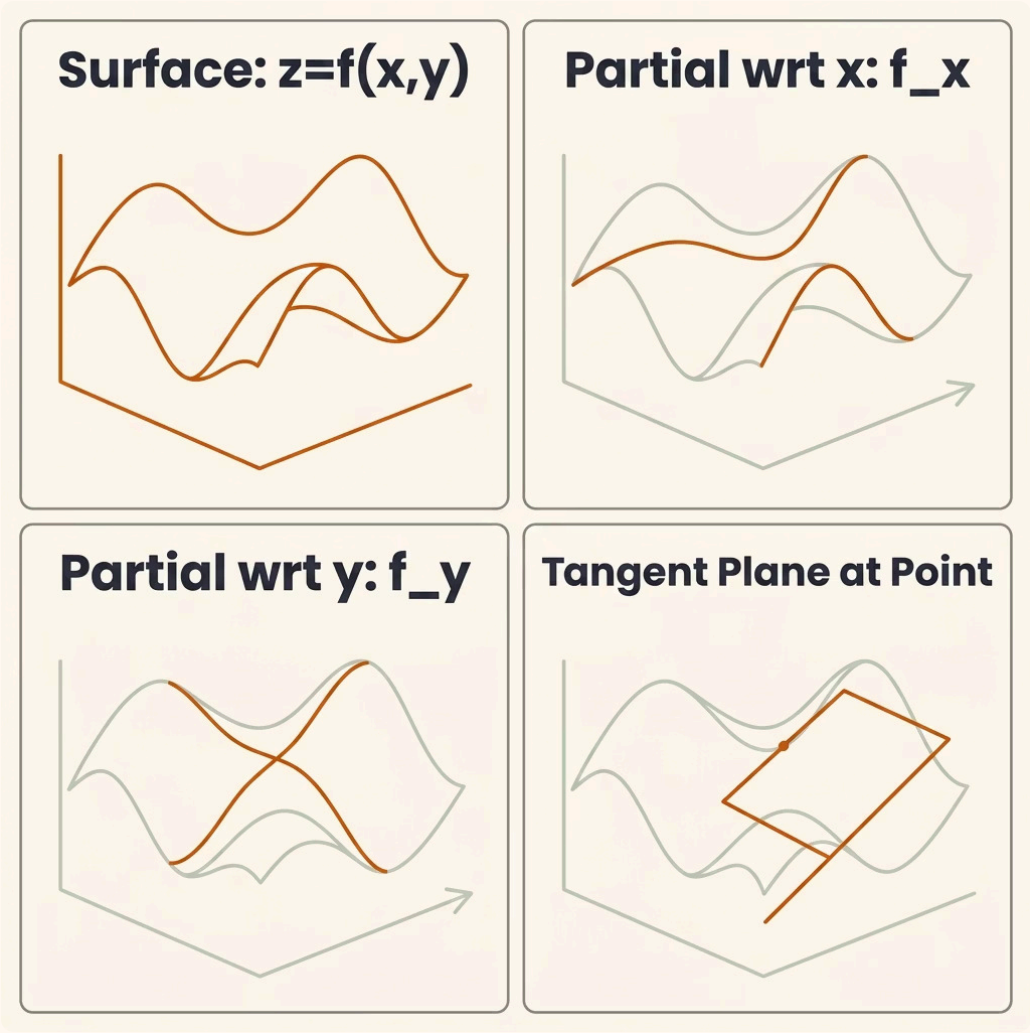
Geometric Interpretation of Partial Derivatives

For a surface $z = f(x, y)$:

- f_x = slope of the curve in the **x-direction** (y fixed)
- f_y = slope of the curve in the **y-direction** (x fixed)
- Together they define the **tangent plane** at a point

If both partials are zero, the tangent plane is **horizontal** — a candidate for maxima or minima.



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$y = x^2$

$\frac{dy}{dx}$

$\frac{d}{dx}(3x+4)$

$y' = 2x$

$y = 3x$

$m = \frac{y_2 - y_1}{x_2 - x_1}$

$\frac{dy}{dx}$

Higher Order Partial Derivatives

Differentiating partial derivatives again yields **second-order** (and higher) derivatives.

Pure Second Derivatives

$f_{xx} = \frac{\partial^2 f}{\partial x^2}$

$f_{yy} = \frac{\partial^2 f}{\partial y^2}$

Mixed Derivatives

$f_{xy} = \frac{\partial^2 f}{\partial y \, \partial x}$

$f_{yx} = \frac{\partial^2 f}{\partial x \, \partial y}$

Clairaut's Theorem

If mixed partials are **continuous**, then $f_{xy} = f_{yx}$ — order of differentiation does not matter.

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The Gradient Vector

The **gradient** collects all first-order partial derivatives into a single vector pointing in the direction of **steepest ascent**.

Visual Insight

On a topographic map, the gradient at any point points **uphill**, perpendicular to contour lines.

Its magnitude $|\nabla f|$ tells you **how steep** the slope is.

Definition

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$$

Key Properties

- Points toward **maximum increase** of f
- Perpendicular to **level curves/surfaces**
- Zero at **critical points** (max, min, saddle)

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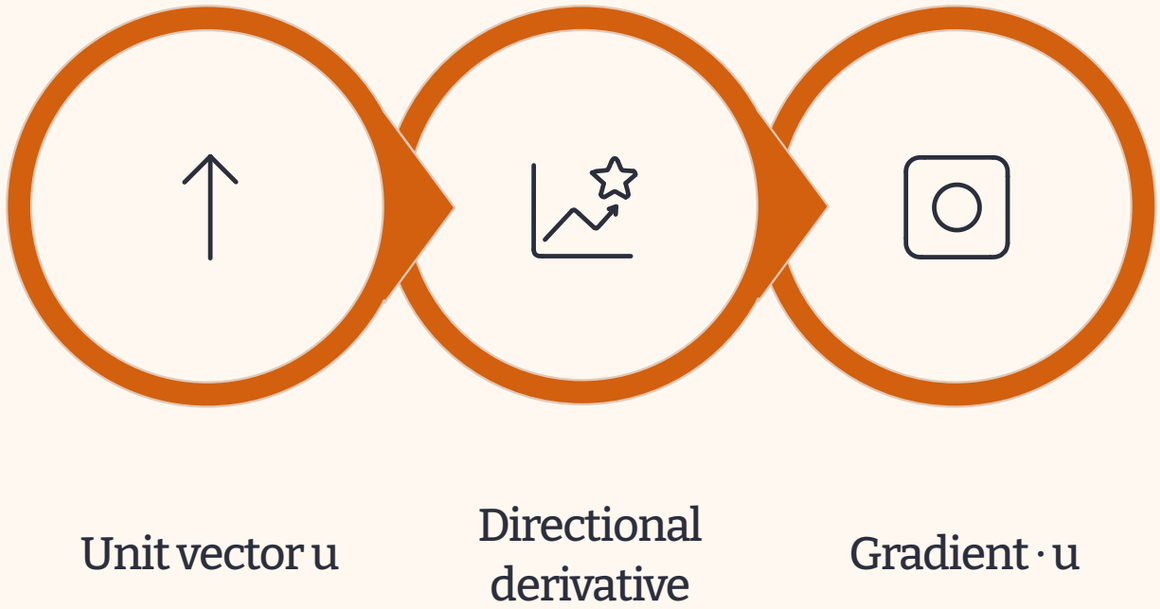
Directional Derivatives

How fast does f change in an **arbitrary direction**? The directional derivative answers this.



The directional derivative is computed as the dot product of the gradient and the unit direction vector.

Formula

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = |\nabla f| \cos\theta$$

where θ is the angle between ∇f and $\mathbf{u}$.

Special Cases

- $\theta = 0°$: maximum increase (along gradient)
- $\theta = 90°$: no change (along level curve)
- $\theta = 180°$: maximum decrease

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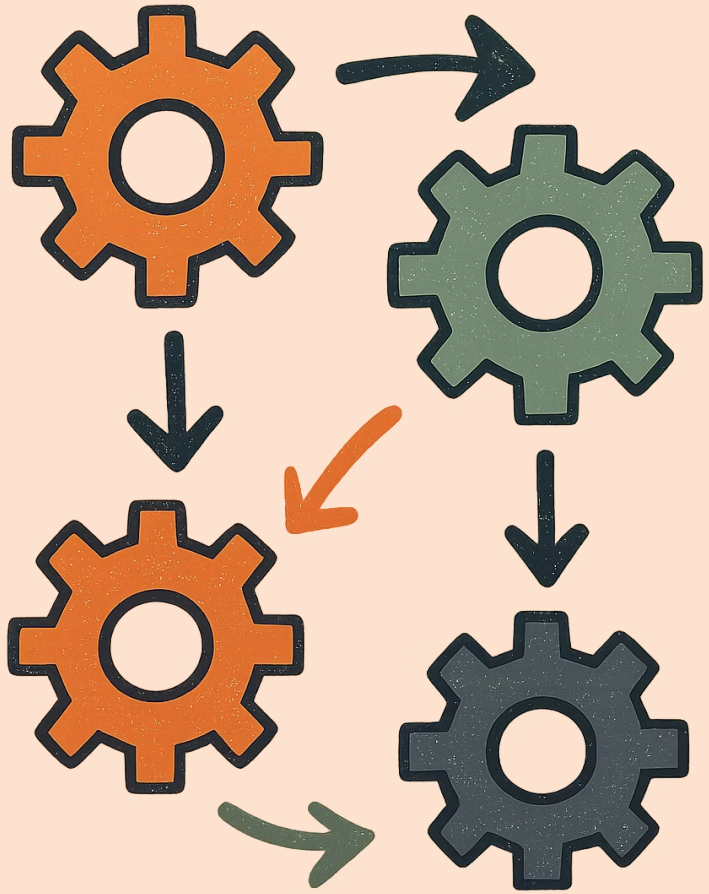

The Chain Rule in Multivariable Calculus

When variables are **interdependent**, the chain rule connects rates of change across nested functions.

1

Single Intermediate

$$z = f(x(t),\, y(t))$$

$$\frac{dz}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt}$$

2

Two Parameters

$$z = f(x(s,t),\, y(s,t))$$

$$\frac{\partial z}{\partial t} = f_x \frac{\partial x}{\partial t} + f_y \frac{\partial y}{\partial t}$$

3

Tree Diagram

Draw a dependency tree; sum all paths from output to input variable.

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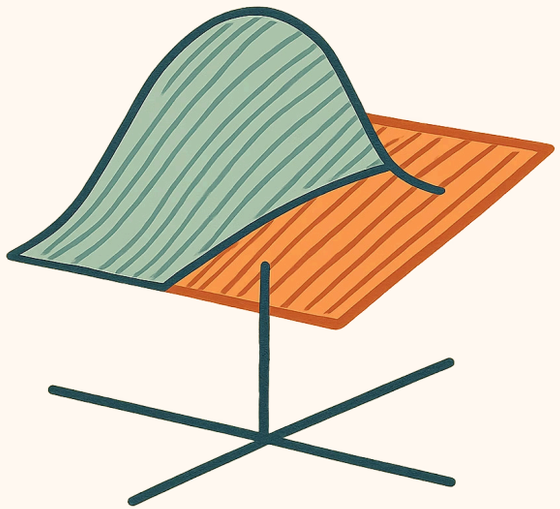
Total Differential and Linear Approximation

The **total differential** estimates how f changes when *all* variables change simultaneously by small amounts.



1

Total Differential

$df = f_x\,dx + f_y\,dy$

2

Linear Approximation

$f(x,y) \approx f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b)$

3

Error Estimation

Used to bound propagated error in measurements and engineering calculations.

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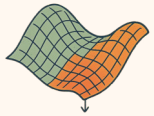
Applications of Multivariable Differentiation

Multivariable differentiation is the mathematical backbone of engineering analysis and design.

Heat & Fluid Flow

Gradient describes heat flux and fluid velocity fields in thermodynamics.

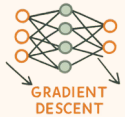


Optimisation

Setting $\nabla f = 0$ finds optimal designs — minimum cost, maximum efficiency.

Robotics & Control

Jacobian matrices (built from partials) map joint velocities to end-effector motion.



Machine Learning

Gradient descent uses ∇f to iteratively minimise loss functions in AI models.

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